REFLECTION CLASSES OF THE EXCEPTIONAL-DIVISOR PERIOD ARE PROPORTIONAL TO THE IRITANI IMAGES OF THE EXCEPTIONAL LINE-BUNDLE DIFFERENCES

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Editorial note. The proof invokes Proposition prop:tw=proj of the same chapter once; it is a navigation stub with its role recorded in the reason, and it is not a bibliography entry. The reflection loops gamma_k(q) referenced in the statement are defined by the author in the immediately preceding discussion of the same section, which the proof text takes as given.

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Allowed standard prerequisites (editorial).

- The period comparison between the twisted exceptional-divisor periods and the projective periods, Proposition prop:tw=proj of this chapter. invoked at proof line 3244. Transports the reflection condition on the twisted period to the proportionality of sigma(beta) with the projective reflection vector.
- The Iritani integral-structure map, whose value on a line-bundle difference is expressed through the Gamma class and the Chern character. used at proof lines 3261-3281. Provides the target expression for the proportionality check: the Gamma class of the blowup factors through the Gamma classes of the base and the exceptional directions, and the Chern character of O((-k+1)E)-O(-kE) contributes the exponential factor.

Source reference. Original proposition internal: sources/sigma-2024-029-reflection-vectors-and-quantum-cohomology-of/pinned-original-source.tex, source lines 3096–3106. The reference is invoked at proof line 3244 to transport the reflection condition to the projective side; it is the author's own preceding result in the same chapter, delivered as its own card, and the proof text here is preserved verbatim as the author wrote it.

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Corollary 5.7. *If $\beta \in \widetilde{H}(E)$ is such that the analytic continuation of ${}^{tw}I^{(-m)}_{\beta}(Q,\lambda)$ along $\gamma_k(q)$ is ${}^{tw}I^{(-m)}_{-\beta}(Q,\lambda)$, then β must be proportional to*

$$\Psi(\mathcal{O}_E(-k+1)) = (2\pi)^{\frac{1-n}{2}}\Gamma(\mathrm{Bl}(X))Q^{-e(n-1)}(2\pi\mathbf{i})^{\deg}\mathrm{ch}(\mathcal{O}_E(-k+1)),$$

where $\mathcal{O}_E(-k+1) := \mathcal{O}(-(k-1)E) - \mathcal{O}(-kE)$.

Proof. According to the above discussion and Proposition internal, under the isomorphism $\widetilde{H}(E) \cong H\big(\mathbb{P}^{n-2}\big)$, $\sigma(\beta)$ must be proportional to $\Psi_q(\mathcal{O}(k))$, that is, β is proportional to

$$e^{-\pi\mathbf{i}\theta}\ (2\pi)^{\frac{3-n}{2}}\Gamma(1+p)^{n-1}q^{-p}e^{2\pi kp} = (2\pi)^{(3-n)/2}\mathbf{i}^{2-n}\Gamma(1-p)^{n-1}q^{p}e^{-2\pi\mathbf{i}kp}.$$

Note that $q^p = e^{\pi\mathbf{i}p}Q^{-p(n-1)}$. Therefore, under the isomorphism $\widetilde{H}(E) \cong H\big(\mathbb{P}^{n-2}\big)$, the above expression becomes

$$(2\pi)^{(3-n)/2}\mathbf{i}^{2-n}\Gamma(1-e)^{n-1}Q^{-e(n-1)}e^{(-2k+1)\pi\mathbf{i}e}e.$$

We have to check that the above expression is proportional to the image of the Iritani map for $\mathrm{Bl}(X)$ of the exceptional object $\mathcal{O}((-k+1)E) - \mathcal{O}(-kE)$. We have

$$\Gamma(\mathrm{Bl}(X)) = \Gamma(X)\Gamma(1-e)^n\Gamma(1+e),$$

$$\Gamma(1-e)\Gamma(1+e) = \frac{2\pi\mathbf{i}e}{e^{\pi\mathbf{i}e} - e^{-\pi\mathbf{i}e}} = \frac{2\pi\mathbf{i}e}{e^{2\pi\mathbf{i}e}-1}e^{\pi\mathbf{i}e},$$

and

$$(2\pi\mathbf{i})^{\deg}\mathrm{ch}(\mathcal{O}((-k+1)E) - \mathcal{O}(-kE)) = e^{-2\pi\mathbf{i}ke}\big(e^{2\pi\mathbf{i}e}-1\big).$$

Since $\Gamma(X)\cup e = e$, the image of the Iritani map becomes

$$(2\pi)^{(3-n)/2}\mathbf{i}\Gamma(1-e)^{n-1}Q^{-e(n-1)}e^{(-2k+1)\pi\mathbf{i}e}e.$$

The claim of the lemma follows. □